

$$1. a. T = 328^{\circ}F \quad v = 1.12 \text{ ft}^3/\text{lb} \quad h = 521 \text{ Btu}/\text{lb}$$

$$b. p = 73.3 \text{ lbf}/\text{in}^2$$

$$v_g = 3.97 \text{ ft}^3/\text{lb} \quad V_g = 569 \text{ Btu}/\text{lb}$$

$$v_f = .0253 \text{ ft}^3/\text{lb} \quad v_f = 86.3 \text{ Btu}/\text{lb}$$

$$v = v_f + x (v_g - v_f)$$

$$= .0253 + .5 (3.97 - .0253)$$

$$= 1.9976 \text{ ft}^3/\text{lb}$$

$$V = v_f + x (V_g - V_f)$$

$$= 86.3 + .5 (569 - 86.3)$$

$$= 327.65 \text{ Btu}/\text{lb}$$

$$c. T = 480^{\circ}C \quad u = 3030 \frac{\text{kJ}}{\text{kg}} \quad h = 3730 \frac{\text{kJ}}{\text{kg}}$$

$$d. p = 1.33 \text{ bar}$$

$$v = .145 \text{ m}^3/\text{kg}$$

$$u = 246 \text{ kJ}/\text{kg}$$

2. at 180° .

$$P_1 = 1.0022 \text{ MPa}$$

$$V_1 = .19405 \text{ m}^3/\text{kg}$$

$$V_2 = V_1$$

$$V_{f2} = 1.0435 \cdot 10^{-3} \text{ m}^3/\text{kg}$$

$$V_{g2} = 1.673 \text{ m}^3/\text{kg}$$

$$P_2 = 1.514 \text{ bar} = 1.514 \text{ MPa}$$

$$V = V_f + x (V_g - V_f)$$
$$x = \frac{.19405 - 1.0435 \cdot 10^{-3}}{1.673 - 1.0435 \cdot 10^{-3}}$$

$$x = .1154$$

$$3, \rho_1 = 300 \text{ kg/m}^3$$

$$V_1 = 2.664 \text{ m}^3$$

$$.6055 \cdot 2 = .1211 \text{ m}^3$$

$$V_1 = .1211 \text{ m}^3$$

$$T_1 = 133.525^\circ\text{C}$$

$$p_2 = 300 \text{ kPa}$$

$$\rho_2 = 1.032 \text{ kg/m}^3$$

$$T_2 = 400.294$$

$$300 \text{ kPa} \cdot (2.664 - .1211)$$

$$300 \text{ kPa} \cdot 1.9429 \text{ m}^3$$

~~$$1.9429 \cdot 300 \text{ kJ}$$~~

80%

$$4. P_1 = 10 \text{ bar} \quad P_2 = 2 \text{ bar}$$

$$T_1 = 66^\circ \text{C} \quad T_2 = 36^\circ \text{C}$$

$$m = 1 \text{ kg} \quad w = 20 \text{ kJ}$$

$$h_1 = 293.38 \text{ kJ/kg} \quad v_1 = 0.023068 \text{ m}^3/\text{kg}$$

$$h_2 = 278.89 \text{ kJ/kg} \quad v_2 = 0.11874 \text{ m}^3/\text{kg}$$

$$9. V_1 = 1 \cdot 0.023068 \text{ m}^3/\text{kg}$$

$$= 0.023068 \text{ m}^3/\text{kg}$$

$$= 23.068 \cdot 10^{-3} \text{ m}^3/\text{kg}$$

$$6. 1 (293.38 - 278.89) + dQ = 20 \text{ kJ}$$

$$dQ = 5.51 \text{ kJ}$$

$$5. \quad T_1 = 70^\circ\text{C}$$

$$T_2 = 120^\circ\text{C}$$

$$V_2 = .89133 \text{ m}^3/\text{kg}$$

$$V_2 = V_g \text{ at } 120 = 2528.9 \text{ kJ/kg}$$

closed so $u_1 = u_2$

$$u_1 = u_f + \left(\frac{u_g - u_f}{v_g - v_f} \right) (v_g - v_f)$$

$$243.04 + \left(\frac{.89133 - .001027}{5.6396 - .001027} \right) (2468.9 - 243.04)$$

$$u_1 = 677.51 \text{ kJ/kg}$$

$$P_1 = P_{\text{sat}}$$

$$P_1 = 31.262 \text{ kPa}$$

$$m(u_2 - u_1) = 2(2528.9 - 677.51)$$

$$? : 3763 \text{ kJ}$$

80%

Try to make sure numbers are the same when using online sources.
I usually change them a bit. It should be 1.5, not 2

$$6. V_f = .8352 \cdot 10^{-3} \text{ m}^3/\text{kg}$$

$$V_g = .0236 \text{ m}^3/\text{kg}$$

$$V = .8352 \cdot 10^{-3} + .6 \cdot (.0236 - .8352 \cdot 10^{-3})$$

$$= .01449 \text{ m}^3/\text{kg}$$

$$.6 = \frac{m_g}{m_g + 25}$$

$$m_g = 37.5 \text{ kg}$$

$$V_g = .0236 \cdot 37.5 = .889 \text{ m}^3$$

$$25 \cdot .8352 \cdot 10^{-3} = .02088 \text{ m}^3$$

Volume of tank

$$V = m \cdot v$$

$$V = 25 + 37.5 \cdot .01449$$

$$= .9056 \text{ m}^3$$

$$\text{Vapor} = \frac{.889}{.9056} \cdot 100$$

$$= 97.725\%$$

$$7. R_{HE} = 2.0785 \frac{\text{kJ}}{\text{kg} \cdot \text{K}}$$

$$P_i V_i = m R_{HE} T_i$$

$$V_i = \frac{m R_{HE} T_i}{P_i}$$

$$V_i = .959 \cdot 10^{-3} \text{ m}^3$$

$$\text{or} \\ 959 \text{ cm}^3$$

$$V_2 = 1.506 \cdot 10^{-3} \text{ m}^3$$

$$\text{or} \\ 1506 \text{ cm}^3$$

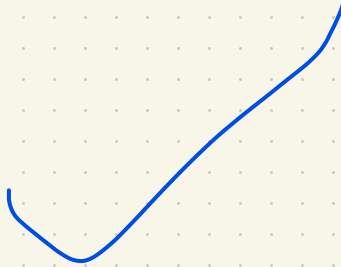
$$8. \quad m = \frac{P \cdot V}{R T_1}$$

$$u_1 = 6889$$

$$u_2 = 14622$$

$$\frac{2.75 \cdot 10^5}{1889.275}$$

$$, 26469 \text{ kg}$$



$$Q = u \cdot w$$

$$w = Q - m(u_2 - u_1) = (-2.75) - (.265) \cdot (14622 - 6889)$$

$$-2653.055 \text{ kJ}$$

Need to divide by Molar Mass.

800 g